Home Lab Project Week 8, Day 2

Task

Map the firewall interfaces to the appropriate VirtualBox networks and verify connectivity between the firewall, Kali Linux, and UTM.

OPNsense First-Boot Console Wizard

Prompt	Answer	Why
Change web GUI protocol from HTTPS to HTTP?	**N** (keep HTTPS)	Keeps management traffic encrypted; no good reason to downgrade
Generate a new self-signed web GUI certificate?	**y** (or leave default)	Needed to support HTTPS; can also be regenerated later from the web GUI under System → Trust → Certificates
Restore web GUI access defaults?	Either is fine on a fresh install	Nothing to restore yet, so no practical effect

Breaking down what falls under Day 2 specifically:

1. **Boot the VM and run the actual installer** — the keymap selection, disk partitioning, and installation to the virtual disk. You did this (you got the login prompt, which only happens after install completes).
2. **First login and initial console setup** — logging in with `root`/`opnsense`, and the three setup-wizard prompts you just answered (HTTPS vs HTTP, certificate generation, restore defaults). This is all part of Day 2's "configure" piece.
3. **Configure WAN and LAN interfaces** — this is explicitly named in Day 2's own task text too ("Configure the required WAN and LAN interfaces"), which actually overlaps with what Day 3 asks for ("Map the firewall interfaces to the appropriate VirtualBox networks and verify connectivity"). Looking at your two task descriptions side by side, Day 2 is about configuring the interfaces logically inside OPNsense (assigning which interface is WAN vs LAN, giving them IPs),

while Day 3 is about making sure those interfaces correctly line up with real VirtualBox networks and that traffic actually flows end-to-end.

So in practice, **Option 1 (Assign Interfaces) and Option 2 (Set IP Address) are Day 2 work**.

Step 1 — Log in

At the console, log in with:

- Username: `root`
- Password: `opnsense`

Step 2 — Option 1: Assign Interfaces

From the main menu, type `1` and Enter. It will walk you through something like:

Valid interfaces are:
vtnet0 ...
vtnet1 ...

Do you want to configure LANs interface(s) now? [y/n]

It'll ask you to type the interface name for **WAN** first, then **LAN**. Since you only have one network adapter on this VM right now (we haven't added a second yet in VirtualBox Settings), you'll only see one interface option — likely `vtnet0`. That's a problem for *finishing* this properly: a firewall needs at least two separate adapters (one for WAN, one for LAN) to make sense of this step.

So before continuing, check: does your OPNsense VM currently have **more than one network adapter** set up in VirtualBox (Settings → Network → Adapter 1, Adapter 2)? If it only has one, Option 1 won't have a second interface to assign as LAN, and we should add that adapter in VirtualBox first, then come back to this.

Step 2 (once you have two+ adapters) — Option 2: Set Interface IP Address

Now let’s configure the ip addresses
We will choose option 2 and hit enter.

- WAN: leave on DHCP (from VirtualBox's NAT), or static if you prefer
- LAN: a static IP in my chosen LAN subnet, e.g. `192.168.78.1/24`

```
 LAN (em0)       -> v4: 192.168.1.1/24
 WAN (em1)       -> v4/DHCP4: 10.0.3.15/24
                    v6/DHCP6: fd17:625c:f037:3:a00:27ff:fe95:80db/64

 HTTPS: SHA256 A9 CE 87 9D 2B 23 B1 52 FF B5 A5 7D FF 3C D6 4F
               7D B3 4D 61 2B 2E A3 8A FF 28 02 EB E0 BD 4E 37
 SSH:   SHA256 96Vtqb0QmFOUQ35q+FmvcI6/Xnl+LPhlCm3LVO8qxNU (ECDSA)
 SSH:   SHA256 3ciI8wjPMdq9DGgYEjOtmprMBroLXhBny8eEf2yt0qs (ED25519)
 SSH:   SHA256 LeA/azGpBJ15oL0mEfXqp8P8jpaaCpKYGcMgmH1qPoY (RSA)

  0) Logout                              7) Ping host
  1) Assign interfaces                   8) Shell
  2) Set interface IP address            9) pfTop
  3) Reset the root password            10) Firewall log
  4) Reset to factory defaults          11) Reload all services
  5) Power off system                   12) Update from console
  6) Reboot system                      13) Restore a backup

Enter an option: 0


FreeBSD/amd64 (OPNsense.internal) (ttyv0)

login: S
```

```
------------------------------------------------

*** OPNsense.internal: OPNsense 26.7 (amd64) ***

 LAN (em0)       -> v4: 192.168.1.1/24
 WAN (em1)       -> v4/DHCP4: 10.0.3.15/24
                    v6/DHCP6: fd17:625c:f037:3:a00:27ff:fe95:80db/64

 HTTPS: SHA256 A9 CE 87 9D 2B 23 B1 52 FF B5 A5 7D FF 3C D6 4F
               7D B3 4D 61 2B 2E A3 8A FF 28 02 EB E0 BD 4E 37
 SSH:   SHA256 96Vtqb0QmFOUQ35q+FmvcI6/Xnl+LPhlCm3LVO8qxNU (ECDSA)
 SSH:   SHA256 3ciI8wjPMdq9DGgYEjOtmprMBroLXhBny8eEf2yt0qs (ED25519)
 SSH:   SHA256 LeA/azGpBJ15oL0mEfXqp8P8jpaaCpKYGcMgmH1qPoY (RSA)

  0) Logout                              7) Ping host
  1) Assign interfaces                   8) Shell
  2) Set interface IP address            9) pfTop
  3) Reset the root password            10) Firewall log
  4) Reset to factory defaults          11) Reload all services
  5) Power off system                   12) Update from console
  6) Reboot system                      13) Restore a backup

Enter an option: 1

Do you want to configure LAGGs now? [y/N]:
```

```
*** OPNsense.internal: OPNsense 26.7 (amd64) ***

 LAN (em0)       -> v4: 192.168.1.1/24
 WAN (em1)       -> v4/DHCP4: 10.0.3.15/24
                    v6/DHCP6: fd17:625c:f037:3:a00:27ff:fe95:80db/64

 HTTPS: SHA256 A9 CE 87 9D 2B 23 B1 52 FF B5 A5 7D FF 3C D6 4F
               7D B3 4D 61 2B 2E A3 8A FF 28 02 EB E0 BD 4E 37
 SSH:   SHA256 96Vtqb0QmFOUQ35q+FmvcI6/Xnl+LPhlCm3LVO8qxNU (ECDSA)
 SSH:   SHA256 3ciI8wjPMdq9DGgYEjOtmprMBroLXhBny8eEf2yt0qs (ED25519)
 SSH:   SHA256 LeA/azGpBJ15oL0mEfXqp8P8jpaaCpKYGcMgmH1qPoY (RSA)

  0) Logout                              7) Ping host
  1) Assign interfaces                   8) Shell
  2) Set interface IP address            9) pfTop
  3) Reset the root password            10) Firewall log
  4) Reset to factory defaults          11) Reload all services
  5) Power off system                   12) Update from console
  6) Reboot system                      13) Restore a backup

Enter an option: 1

Do you want to configure LAGGs now? [y/N]: n
Do you want to configure VLANs now? [y/N]: n
```

```
 SSH:   SHA256 LeA/azGpBJ15oL0mEfXqp8P8jpaaCpKYGcMgmH1qPoY (RSA)

  0) Logout                              7) Ping host
  1) Assign interfaces                   8) Shell
  2) Set interface IP address            9) pfTop
  3) Reset the root password            10) Firewall log
  4) Reset to factory defaults          11) Reload all services
  5) Power off system                   12) Update from console
  6) Reboot system                      13) Restore a backup

Enter an option: 1

Do you want to configure LAGGs now? [y/N]: n
Do you want to configure VLANs now? [y/N]: n

Valid interfaces are:

em0             08:00:27:83:07:57 Intel(R) Legacy PRO/1000 MT 82540EM
em1             08:00:27:95:80:db Intel(R) Legacy PRO/1000 MT 82540EM

If you do not know the names of your interfaces, you may choose to use
auto-detection. In that case, disconnect all interfaces now before
hitting 'a' to initiate auto detection.

Enter the WAN interface name or 'a' for auto-detection: em1
```

```
  2) Set interface IP address            9) pfTop
  3) Reset the root password            10) Firewall log
  4) Reset to factory defaults          11) Reload all services
  5) Power off system                   12) Update from console
  6) Reboot system                      13) Restore a backup

Enter an option: 1

Do you want to configure LAGGs now? [y/N]: n
Do you want to configure VLANs now? [y/N]: n

Valid interfaces are:

em0             08:00:27:83:07:57 Intel(R) Legacy PRO/1000 MT 82540EM
em1             08:00:27:95:80:db Intel(R) Legacy PRO/1000 MT 82540EM

If you do not know the names of your interfaces, you may choose to use
auto-detection. In that case, disconnect all interfaces now before
hitting 'a' to initiate auto detection.

Enter the WAN interface name or 'a' for auto-detection: em1

Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(or nothing if finished): em0
```

```
  5) Power off system                   12) Update from console
  6) Reboot system                      13) Restore a backup

Enter an option: 1

Do you want to configure LAGGs now? [y/N]: n
Do you want to configure VLANs now? [y/N]: n

Valid interfaces are:

em0             08:00:27:83:07:57 Intel(R) Legacy PRO/1000 MT 82540EM
em1             08:00:27:95:80:db Intel(R) Legacy PRO/1000 MT 82540EM

If you do not know the names of your interfaces, you may choose to use
auto-detection. In that case, disconnect all interfaces now before
hitting 'a' to initiate auto detection.

Enter the WAN interface name or 'a' for auto-detection: em1

Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(or nothing if finished): em0

Enter the Optional interface 1 name or 'a' for auto-detection
(or nothing if finished):
```

Just hit enter

```
Valid interfaces are:

em0             08:00:27:83:07:57 Intel(R) Legacy PRO/1000 MT 82540EM
em1             08:00:27:95:80:db Intel(R) Legacy PRO/1000 MT 82540EM

If you do not know the names of your interfaces, you may choose to use
auto-detection. In that case, disconnect all interfaces now before
hitting 'a' to initiate auto detection.

Enter the WAN interface name or 'a' for auto-detection: em1

Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(or nothing if finished): em0

Enter the Optional interface 1 name or 'a' for auto-detection
(or nothing if finished):

The interfaces will be assigned as follows:

WAN  -> em1
LAN  -> em0

Do you want to proceed? [y/N]: y
```

Do you want to proceed, type y, and press enter.

The system will configure the interfaces.

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```
 WAN (em1)       -> v4/DHCP4: 10.0.3.15/24
                    v6/DHCP6: fd17:625c:f037:3:a00:27ff:fe95:80db/64

 HTTPS: SHA256 A9 CE 87 9D 2B 23 B1 52 FF B5 A5 7D FF 3C D6 4F
               7D B3 4D 61 2B 2E A3 8A FF 28 02 EB E0 BD 4E 37
 SSH:   SHA256 96VtqbOQmFOUQ35q+FmvcI6/Xnl+LPhlCm3LVO8qxNU (ECDSA)
 SSH:   SHA256 3ciI8wjPMdq9DGgYEjOtmprMBroLXhBny8eEf2yt0qs (ED25519)
 SSH:   SHA256 LeA/azGpBJl5oL0mEfXqp8P8jpaaCpKYGcMgmH1qPoY (RSA)

  0) Logout                              7) Ping host
  1) Assign interfaces                   8) Shell
  2) Set interface IP address            9) pfTop
  3) Reset the root password            10) Firewall log
  4) Reset to factory defaults          11) Reload all services
  5) Power off system                   12) Update from console
  6) Reboot system                      13) Restore a backup

Enter an option: 2

Available interfaces:

1 - LAN (em0 - static, idassoc6)
2 - WAN (em1 - dhcp, dhcp6)

Enter the number of the interface to configure: 1
```

We will answer no to configure IPv4 address lan Interface via DHCP

```
HTTPS: SHA256 A9 CE 87 9D 2B 23 B1 52 FF B5 A5 7D FF 3C D6 4F
              7D B3 4D 61 2B 2E A3 8A FF 28 02 EB E0 BD 4E 37
SSH:   SHA256 96Vtqb0QmFOUQ35q+FmvcI6/Xnl+LPhlCm3LVO8qxNU (ECDSA)
SSH:   SHA256 3ciI8wjPMdq9DGgYEjOtmprMBroLXhBny8eEf2yt0qs (ED25519)
SSH:   SHA256 LeA/azGpBJl5oL0mEfXqp8P8jpaaCpKYGcMgmH1qPoY (RSA)

 0) Logout                              7) Ping host
 1) Assign interfaces                   8) Shell
 2) Set interface IP address            9) pfTop
 3) Reset the root password            10) Firewall log
 4) Reset to factory defaults          11) Reload all services
 5) Power off system                   12) Update from console
 6) Reboot system                      13) Restore a backup

Enter an option: 2

Available interfaces:

1 - LAN (em0 - static, idassoc6)
2 - WAN (em1 - dhcp, dhcp6)

Enter the number of the interface to configure: 1

Configure IPv4 address LAN interface via DHCP? [y/N] n
```

```
SSH:   SHA256 96Vtqb0QmFOUQ35q+FmvcI6/Xnl+LPhlCm3LVO8qxNU (ECDSA)
SSH:   SHA256 3ciI8wjPMdq9DGgYEjOtmprMBroLXhBny8eEf2yt0qs (ED25519)
SSH:   SHA256 LeA/azGpBJl5oL0mEfXqp8P8jpaaCpKYGcMgmH1qPoY (RSA)

 0) Logout                              7) Ping host
 1) Assign interfaces                   8) Shell
 2) Set interface IP address            9) pfTop
 3) Reset the root password            10) Firewall log
 4) Reset to factory defaults          11) Reload all services
 5) Power off system                   12) Update from console
 6) Reboot system                      13) Restore a backup

Enter an option: 2

Available interfaces:

1 - LAN (em0 - static, idassoc6)
2 - WAN (em1 - dhcp, dhcp6)

Enter the number of the interface to configure: 1

Configure IPv4 address LAN interface via DHCP? [y/N] n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.78.20
```

```
  5) Power off system                   12) Update from console
  6) Reboot system                      13) Restore a backup

Enter an option: 2

Available interfaces:

1 - LAN (em0 - static, idassoc6)
2 - WAN (em1 - dhcp, dhcp6)

Enter the number of the interface to configure: 1

Configure IPv4 address LAN interface via DHCP? [y/N] n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.78.20


Subnet masks are entered as bit counts (like CIDR notation).
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24
```

```
Available interfaces:

1 - LAN (em0 - static, idassoc6)
2 - WAN (em1 - dhcp, dhcp6)

Enter the number of the interface to configure: 1

Configure IPv4 address LAN interface via DHCP? [y/N] n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.78.20


Subnet masks are entered as bit counts (like CIDR notation).
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>
```

Just press enter for the upstream gateway address question.

```
1 - LAN (em0 - static, idassoc6)
2 - WAN (em1 - dhcp, dhcp6)

Enter the number of the interface to configure: 1

Configure IPv4 address LAN interface via DHCP? [y/N] n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.78.20


Subnet masks are entered as bit counts (like CIDR notation).
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via WAN tracking? [Y/n]
```

Just say no to configure IPv6

```
1 - LAN (em0 - static, idassoc6)
2 - WAN (em1 - dhcp, dhcp6)

Enter the number of the interface to configure: 1

Configure IPv4 address LAN interface via DHCP? [y/N] n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.78.20


Subnet masks are entered as bit counts (like CIDR notation).
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via WAN tracking? [Y/n] n
Configure IPv6 address LAN interface via DHCP6? [y/N] n
```

```
Enter the number of the interface to configure: 1

Configure IPv4 address LAN interface via DHCP? [y/N] n

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.78.20


Subnet masks are entered as bit counts (like CIDR notation).
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via WAN tracking? [Y/n] n
Configure IPv6 address LAN interface via DHCP6? [y/N] n

Enter the new LAN IPv6 address. Press <ENTER> for none:
> n
```

```
> 192.168.78.20


Subnet masks are entered as bit counts (like CIDR notation).
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via WAN tracking? [Y/n] n
Configure IPv6 address LAN interface via DHCP6? [y/N] n

Enter the new LAN IPv6 address. Press <ENTER> for none:
> n

Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? [y/N] y
```

```
Subnet masks are entered as bit counts (like CIDR notation).
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via WAN tracking? [Y/n] n
Configure IPv6 address LAN interface via DHCP6? [y/N] n

Enter the new LAN IPv6 address. Press <ENTER> for none:
> n

Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? [y/N] y

Enter the start address of the IPv4 client address range: 192.168.78.1
Enter the end address of the IPv4 client address range: 192.168.78.19
```

```
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via WAN tracking? [Y/n] n
Configure IPv6 address LAN interface via DHCP6? [y/N] n

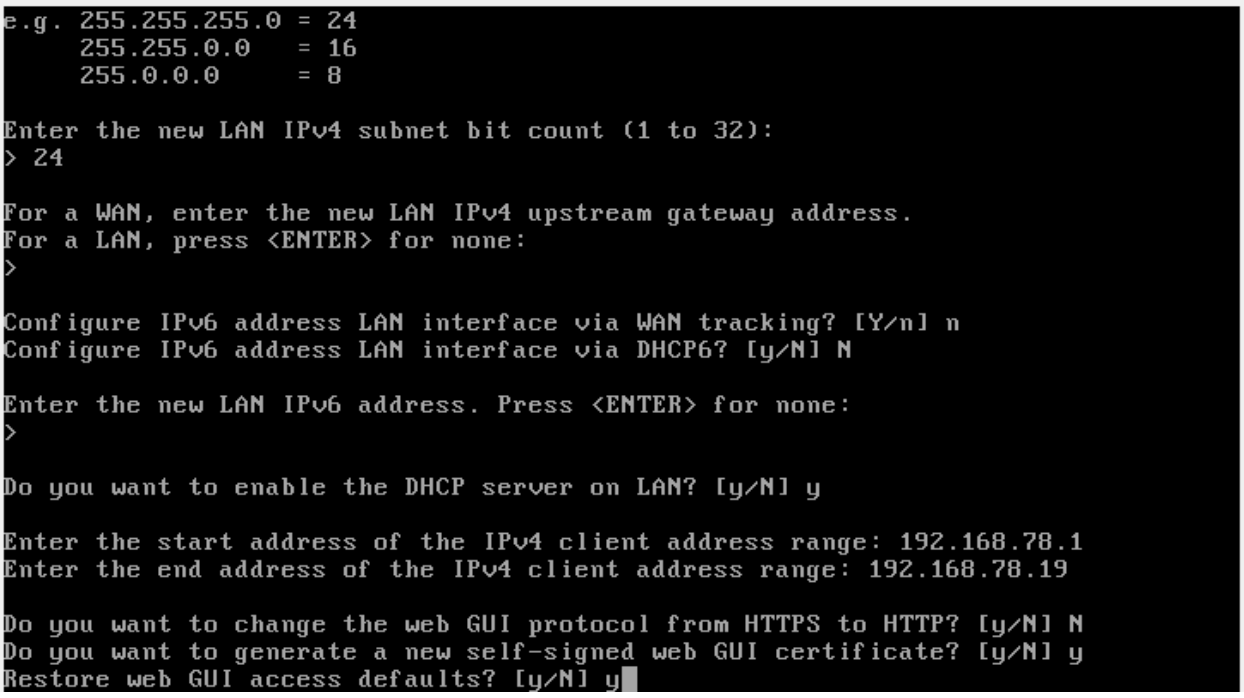
Enter the new LAN IPv6 address. Press <ENTER> for none:
> n

Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? [y/N] y

Enter the start address of the IPv4 client address range: 192.168.78.1
Enter the end address of the IPv4 client address range: 192.168.78.19

Do you want to change the web GUI protocol from HTTPS to HTTP? [y/N] N
```


e.g. 255.255.255.0 = 24
255.255.0.0 = 16
255.0.0.0 = 8
Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24
For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>
Configure IPv6 address LAN interface via WAN tracking? [Y/n] n
Configure IPv6 address LAN interface via DHCP6? [y/N] N
Enter the new LAN IPv6 address. Press <ENTER> for none:
>
Do you want to enable the DHCP server on LAN? [y/N] y
Enter the start address of the IPv4 client address range: 192.168.78.1
Enter the end address of the IPv4 client address range: 192.168.78.19
Do you want to change the web GUI protocol from HTTPS to HTTP? [y/N] N
Do you want to generate a new self-signed web GUI certificate? [y/N] y
Restore web GUI access defaults? [y/N] y